

ALOK RANJAN TRIPATHY

B.Tech in Computer Science and Engineering

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Education

B.Tech in Computer Science and Engineering, IIIT Bhubaneswar 2028
Expected Graduation: 2028

Skills

Languages

Python, C, C++

Tools

Git, Jupyter Notebook, Streamlit, Flask

Frameworks & Libraries

Machine Learning:

PyTorch, TensorFlow, scikit-learn

Computer Vision:

OpenCV, MediaPipe, YOLO, Segment Anything (SAM)

LLM & GenAI:

LangChain, LangGraph

Projects

Sunglasses Transfer via Segmentation 📁, SAM + YOLO + OpenCV

Built a vision pipeline to segment, extract, and overlay sunglasses between faces in real-time. Combined SAM for segmentation, YOLO for detection, and MediaPipe for landmark alignment to ensure realistic placement and scaling.

Handwritten Digit Recognition using ANN from Scratch using NumPy

Implemented a fully connected Artificial Neural Network from scratch using NumPy for handwritten digit classification on the MNIST dataset. Developed forward propagation, backpropagation, gradient computation, and parameter updates without deep learning frameworks. Trained and evaluated the model while exploring the effects of architecture and hyperparameter choices on model performance.

Spell Gesture Recognition 📁, LSTM, OpenCV, MediaPipe

Developed a real-time gesture recognition system inspired by "Harry Potter" spell casting. Used OpenCV and MediaPipe for motion tracking and trained a custom LSTM network for recognizing user-defined gestures. Enabled dynamic class creation, retraining, and storage for personalized gestures.

RAG based LLM Chatbot 📁, LangChain, Mistral

Built a context-aware document question-answering system using Retrieval-Augmented Generation (RAG). Designed the pipeline for PDF parsing, chunk generation, embedding-based retrieval, and context injection into a Large Language Model. Improved response relevance through semantic search and retrieval-based grounding, enabling accurate information extraction from unstructured documents.

Certifications & Achievements

AI & Data Analytics Virtual

Experience

- Tata iQ, Forage
- 2025